

FAST GENERATION SHEDDING EQUIPMENT BASED ON THE OBSERVATION OF SWINGS OF GENERATORS

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Abstract - This paper presents a new method and newly developed stabilizing equipment to prevent the loss of synchronism of generators in pumped-storage plants from spreading. The proposed method includes functions to estimate the swing of each generator by using on-line generator output sampled at a rate of 600 times per second after an occurrence of disturbance such as fault, faulty equipment and so on, to predict generator swing 200-300 milliseconds ahead and loss of synchronism between generators in pumped-storage plants and those in thermal and nuclear plants, and to decide the number of generators that must be shed to maintain stability.

Tokyo Electric Power Co. (TEPCO) installed the newly developed stabilizing equipment based on this method in the power system in June, 1986.

1. INTRODUCTION

The relative phase angle differences between the generators in pumped-storage plants and the generators in nuclear and thermal plants are exaggerated when generators in the pumped-storage plants operate in pumping mode. If a disturbance occurs in this operating condition, generators in the pumped-storage plant may easily lose synchronism with the other generators.

In an actual power system, various contingencies may occur. Therefore, depending on the severity of the contingencies, one generator or many may lose synchronism. Especially when there are several pumped-storage plants in the power system, generators in single or multiple pumped-storage plants may lose synchronism with the rest of the system while those in nuclear thermal plants remain in a coherent group. As a result, the frequency of the system will move out of the allowable range.

In Japan, some equipments for stabilization have been installed in actual power systems. [1] But the equipment in each system has been installed at a specific generating plant which may lose synchronism due to a fault occurring at a specific line connected to the plant. Also, the prediction of step-out and the determination of the amount of generation shedding necessary for stabilization are based on the value of the generator output observed at a steady state and the kinds of fault detected. Theoretical studies on transient stability assessment and stabilization have been widely conducted. Most of them are based on transient energy function. [2-6]

To cope with above-mentioned instabilities, a new approach based on shedding the units in the pumped-storage plants is proposed. In addition, the new stabilizing equipment developed is described.

The stabilizing method proposed has several functions. The first function is to estimate the angular velocity and phase angle of a generator by using generator output sampled at a rate of 600 times per second for about one second after a disturbance occurs. The second function is to predict the change of phase angle 200-300 milliseconds ahead and the loss of synchronism of generators in pumped-storage plants with those in nuclear thermal plants by comparing their phase angles. The third function is to determine the number of units in the pumped-storage plant to be shed to maintain system stability.

2. MODES OF INSTABILITY IN TEPCO SYSTEM AND STABILIZING METHOD

TEPCO's power system, shown in Fig.1, comprises 30 generating plants including eight pumped-storage plants. Total generating capacity of the pumped-storage plants is about ten percent of the total generating capacity of the system.

Many factors, such as fault location, fault duration and kinds of fault, affect generator swing. For example, a single pumped-storage plant may lose synchronism due to a disturbance, but many plants may be stepped out by other disturbances. If a disturbance occurs at some place in the network when pumped-storage plants are in pumping operation, the generators in thermal and nuclear plants are accelerated during the disturbance, but the generators in the pumped-storage plants are decelerated. After the fault is cleared, generators in nuclear and thermal plants swing together but the relative phase angles between the pumped-storage generators and thermal nuclear generators are exaggerated and some generators may lose synchronism.

When thermal and nuclear generators are in a coherent group, one generator can be selected as representative of all the generators in that group and termed the reference generator.

The observation of the relative swings between the generators of each pumped-storage plant and the reference generator is considered to be useful for prediction of loss of synchronism. From now on, the units in each pumped storage plant are expressed as a single equivalent generator.

The validity of generation shedding can be explained as follows. When units are shed, the load on the equivalent generator is reduced, but the amount of the reduction in power received from the network is smaller than the reduction of the load, because the phase angle does not change when generation shedding takes place. Therefore, accelerating power to the equivalent generator increases and, as a result, the relative phase angle difference becomes smaller and returns to the stable state.

A general configuration of stabilizing equipment is shown in Fig.2. At steady state, generator output power, current and voltage at the generator

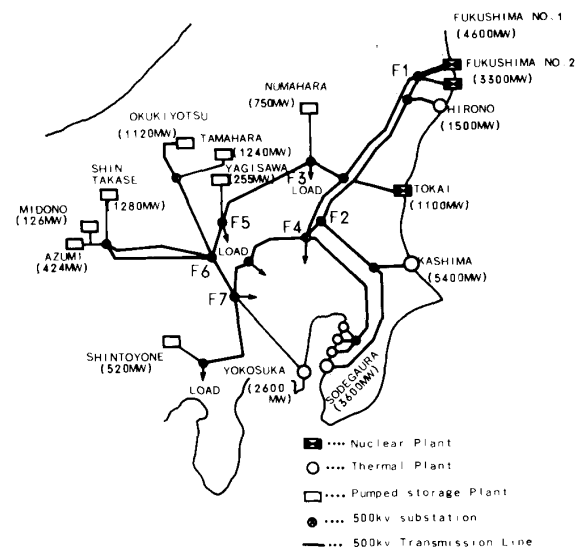


Fig.1 GENERAL CONFIGURATION OF TEPCO POWER SYSTEM

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terminal are measured, and are transmitted to a plant processor installed at pumped-storage plant over a microwave channel. The output power is used as mechanical input power. Relative phase angle between the reference generator and the generator in the pumped-storage plant is obtained by using the measured terminal voltage phase angles. The internal phase angle between the internal voltage and the terminal voltage in each generator is also computed by using the measured generator current, the internal impedance and also the measured terminal voltage. When a rapid change of generator output caused by disturbance is detected, the voltage, current, and output power of the reference generator and the pumped-storage generator are sampled at a rate of 600 times per second. The data are transmitted to the plant processor. The plant processor estimates angular velocity and phase angle by using the data, predicts the loss of synchronism, decides the number of units to be shed and sends a command to drop units. In order to make the system highly reliable, one plant processor is installed at each pumped-storage plant.

3. ALGORITHM

The proposed method is explained in detail by using the procedure shown in Fig.3

3.1 Estimation of Swing of Generator

We set the length of the time interval equal to one cycle period (20ms). Since generator output power is sampled at a rate of 600 times per second, 12 output powers can be measured at each interval. Then, the average output power P_K at each interval is computed and used as the value of the generator output power in that interval. This processing reduces the effect of noise on measurement. Since mechanical input P_{IN} is almost constant for about one second after a disturbance occurrence, P_{IN} is used as a constant value. Then the incremental value $\Delta\omega_K$ of angular velocity during interval k is calculated by eq.(1)

$$\Delta\omega_K = (P_{IN} - P_K) \Delta t / M \quad (1)$$

Angular velocity ω_K and phase angle δ_K are calculated by eq. (2) and (3)

$$\omega_K = \omega_{K-1} + \Delta\omega_K \quad (2)$$

$$\delta_K = \delta_{K-1} + 0.5(\omega_K + \omega_{K-1})\Delta t \quad (3)$$

where,

$$\omega_0 = 0, \quad \delta_0 = 0$$

3.2 Prediction of Loss of Synchronism

Angular velocity ω varies continuously and shows a smooth change because of the large moment of inertia of the generator. In order to predict angular velocity ω and phase angle δ , a new function extrapolating the three points $\omega(r_0)$, $\omega(r_1)$, $\omega(r_2)$, shown in Fig.4, is defined in eq. (4)

$$\omega(t) = \omega(r_2) + a_1(t-r_2) + a_2(t-r_1)(t-r_2) \quad (4)$$

where

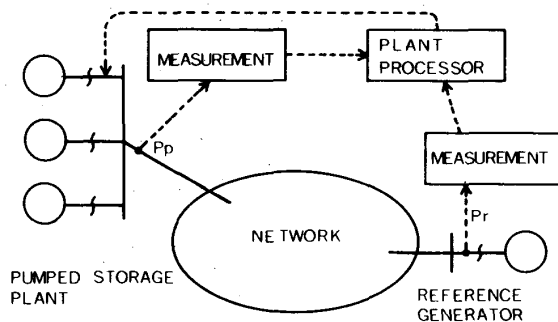


Fig.2 GENERAL CONFIGURATION OF STABILIZING EQUIPMENT

$$\left. \begin{aligned} a_1 &= (\omega(r_2) - \omega(r_1)) / (r_2 - r_1) \\ a_0 &= (\omega(r_1) - \omega(r_0)) / (r_1 - r_0) \\ a_2 &= (a_1 - a_0) / (r_2 - r_0) \end{aligned} \right\} \quad (5)$$

In addition, a function predicting the phase angle is expressed by eq. (6) which is defined by integrating eq. (4).

$$\begin{aligned} \delta(t) &= \delta(r_0) + \omega(r_2)(t-r_0) + a_1((t^2-r_0^2)/2 \\ &\quad - r_2(t-r_0)) + a_2((t^3-r_0^3)/3 \\ &\quad - (r_1+r_2)(t^2-r_0^2)/2 + r_1 r_2(t-r_0)) \end{aligned} \quad (6)$$

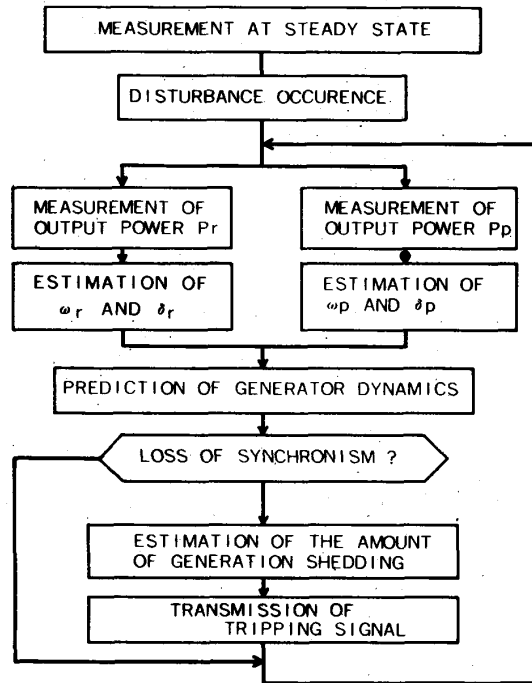


Fig.3 PROCEDURE FOR STABILIZATION

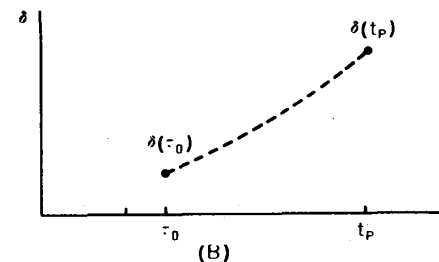
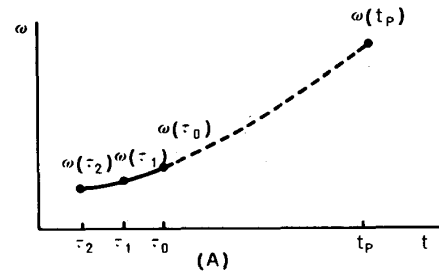


Fig.4 PREDICTION OF ω AND δ

By using above the equations, the phase angles of the reference generator and the generator in the pumped-storage plant can be predicted.

The relative phase angle $\Delta\delta_{pr}(t)$ between two generators is defined as follows

$$\Delta\delta_{pr}(t) = \delta r(t) - \delta p(t) + \Delta\delta_{pr}(0) \quad (7)$$

Then, if $\Delta\delta_{pr}(t)$ shows a tendency to increase and exceeds the prespecified threshold at a specified time in the future, a loss of synchronism can be predicted. In eq. (7), $\Delta\delta_{pr}(0)$ is a relative phase angles between internal voltages at both generators, which are observed in the steady state. The threshold is decided by using the results of many simulation studies.

3.3 Estimation of Generation-Shedding

To decide the number of units to be shed for stabilization requires:

- A method to estimate the output power of the pumped-storage plant immediately after a given number of units of the plant are shed.
- An algorithm to estimate the change in phase angle caused by generation-shedding.

(1) Estimation of Output Power after Generation-Shedding

An equation to estimate the generator output power, when m units out of n in the pumped-storage plant are shed, is derived.

The changes in the voltage magnitude and swing of other generators due to partial shedding of generation in a pumped-storage plant is very small compared with the total generating capacity in the power system.

Therefore, the part of the network seen from the main transformer in the pumped-storage plant is represented by driving point impedance $\hat{Z}_D$ which is connected to the infinite bus as shown in Fig.5. G_1 is the equivalent generator composed of $(n-m)$ units and G_2 is the one composed of the m units to be shed. X_G is the summation of internal impedance X_d'' and the impedance of transformer X_t . Then $\hat{Z}_1$ and $\hat{Z}_2$ are expressed as

$$\hat{Z}_1 = jnX_G / (n-m) \quad (8)$$

$$\hat{Z}_2 = jnX_G / m \quad (9)$$

The relation between internal voltages $\hat{e}_1''$ and $\hat{e}_2''$ and generator currents $\hat{I}_1$ and $\hat{I}_2$ are expressed by the eq.(10)

$$\begin{bmatrix} \hat{e}_1'' \\ \hat{e}_2'' \end{bmatrix} = \begin{bmatrix} \hat{Z}_1 + \hat{Z}_D & \hat{Z}_D \\ \hat{Z}_D & \hat{Z}_2 + \hat{Z}_D \end{bmatrix} \begin{bmatrix} \hat{I}_1 \\ \hat{I}_2 \end{bmatrix} + \begin{bmatrix} \hat{E} \\ \hat{E} \end{bmatrix} \quad (10)$$

Since the parameters of every generator in the pumped-storage plant are the same, the internal voltages at both generators are equal, as shown in eq.(11).

$$\hat{e}_1'' = \hat{e}_2'' \quad (11)$$

Then the total generator output power P can be computed by eq.(12).

$$P = \text{Real} \left[\frac{\hat{Z}_2 (|\hat{e}_1''|^2 - \hat{e}_1'' \hat{E})}{\hat{Z}_1 (\hat{Z}_1 + \hat{Z}_2) + \hat{Z}_1 \hat{Z}_2} \right] \frac{n}{n-m} \quad (12)$$

In addition, since voltage $\hat{e}_1''$ does not change in the instants, before and immediately after the generation shedding, the generator output power P^* of the $(n-m)$ units is computed by eq.(13)

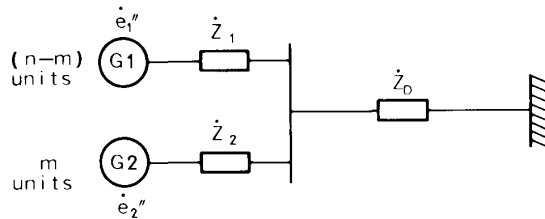


Fig.5 EQUIVALENT NETWORK

$$P^* = \text{Real} \left[\frac{(\hat{e}_1''|^2 - \hat{e}_1'' \hat{E})}{(\hat{Z}_1 + \hat{Z}_D)} \right] \quad (13)$$

As the real part of the driving point impedance r is very small compared with the imaginary part of the impedance X_D , $\hat{Z}_D$ can be expressed approximately by

$$\hat{Z}_D = jX_D \quad (14)$$

Then, P^* can be calculated by using P , and constants X_G , X_D , m , and n .

$$P^* = \frac{X_G + X_D}{X_G + \frac{n-m}{n} X_D} \cdot \frac{n-m}{n} P = K \cdot P \quad (15)$$

(2) Estimation of the Amount of Generation-shedding

Suppose the predicted relative phase angle $\Delta\delta_{su}(T_p)$ at time T_p in the future exceeds the threshold as shown in Fig.6. The relative phase angle is recovered by $\Delta\delta(T_p)$ when m units out of n in the pumped-storage plant are shed at time T_c , and the angle is estimated to be $\Delta\delta_{su}^*(T_p)$. Since the duration time between T_0 and T_p is about 0.25 seconds, generator output power can be assumed to be equal to $P(T_0)$ because the change in output power is very small in that duration time. Therefore, the change of phase angle of the generator is predicted by

$$\Delta\delta_1 = (PIN - P(T_0))(T_p - T_0)^2 / 2M \quad (16)$$

Due to the shedding of m units out of n in the plant, the inertia and mechanical input of the equivalent generator reduces to

$$M^* = M(n-m) / n \quad (17)$$

$$PIN^* = PIN(n-m) / n \quad (18)$$

Also generator output power after the shedding of m units can be predicted by using the constant K derived in eq.(15)

$$P^* = K \cdot P(T_0) \quad (19)$$

Therefore the increased value $\Delta\delta_2$ of the phase angle between T_c and T_p due to partial generation shedding can be predicted approximately by eq.(20)

$$\Delta\delta_2 = (PIN^* - P^*)(T_p - T_c)^2 / 2M^* \quad (20)$$

The angles $\Delta\delta(T_p)$ recovered by the generation shedding can be estimated to be

$$\Delta\delta(T_p) = \Delta\delta_1 - \Delta\delta_2 \quad (21)$$

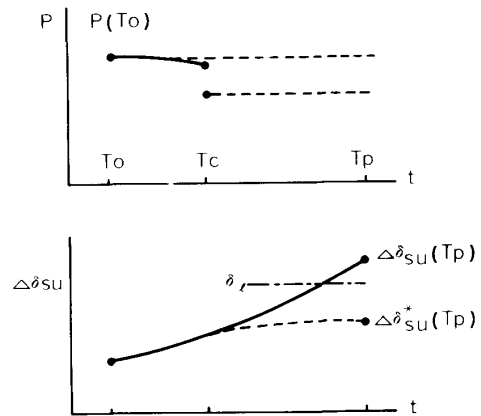


Fig.6 PREDICTION OF ANGLE

Therefore, the predicted $\Delta \delta_{SU}^*(TP)$ can be expressed by the following equation.

$$\Delta \delta_{SU}^*(TP) = \Delta \delta_{SU}(TP) + \Delta \delta_1 - \Delta \delta_2 \quad (22)$$

Using the above procedure, the number of units m to be shed is decided so that following relation is satisfied

$$\Delta \delta_{SU}^*(TP) < \delta_g \quad (23)$$

δ_g is the threshold.

At each pumped-storage plant, one plant processor is used. Therefore, the different thresholds at the different processors are used in order to prevent over-shedding of units.

4. SYSTEM CONFIGURATION

Fig.7 shows the configuration of the stabilizing equipment newly developed based on the proposed method. It consists of microprocessor-based control units (plant processor) and data acquisition units. The plant processor is installed at each pumped-storage plant and perform stabilizing control calculations such as the estimation of generator dynamics, the prediction of loss of synchronism and the estimation of the number of generators to be shed. On the other hand, data acquisition units are installed at each pumped-storage plant and reference generator and measure the output power, terminal voltage and generator current at a sampling rate of 600 HZ.

Data measured at the reference generator is sent to each plant processor over a high-speed (54Kbit per second) digital microwave carrier system. To enhance reliability, the stabilizing equipment is completely duplicated.

Since the plant processor use a 16-bit microprocessor, to enhance precision, calculation is done by double precision. Furthermore, generator shedding is performed only upon reception of both a tripping signal from the plant processor and one from a fail-safe relay (Power decrease detection relay) to prevent incorrect operation of the stabilizing equipment.

Fig.8 shows the front view of the stabilizing equipment installed at pumped-storage plant.

5. VERIFICATION OF EFFECT OF PROPOSED METHOD

5.1 Digital Simulation

(Example 1) Digital simulation using small system

Fig. 9 shows the configuration of the small power system, its line impedances and generation/load pattern, where generators G1, G2 and G3 are pumped-storage plants operating in the pumping mode. These generators represent the equivalent generators consisting of 4, 6, and 2 units, respectively. In Table 1, various fault locations evaluated for the system in Fig.9 are shown. The fault is assumed to occur at terminal NT of the branch between nodes NF and NT at $t_1=0.03$ (sec.) and it is cleared at $t_2=0.10$ (sec.)

Depending on the generation/load patterns and fault locations, various modes of swings appear.

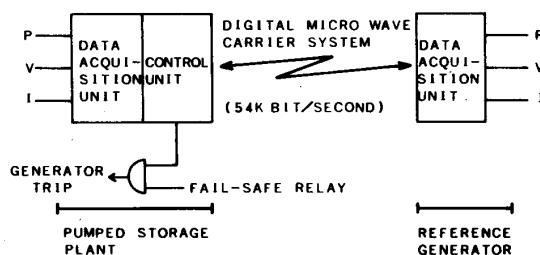


Fig.7 CONFIGURATION OF STABILIZING EQUIPMENT

(1) Validity of angle prediction

To show the accuracy of the angle prediction method mentioned in section 3.2, we compare the predicted values with the exactly calculated ones. Fig.10 shows the errors, when times for prediction are set to 150, 200, 250, 300, (ms), respectively. From Fig.10 it is verified that the prediction errors for a prediction time at less than 250 (ms) are as small as 3 degrees. These results show that the angle prediction method is accurate enough to be used with the stabilizing equipment.

(2) Validity of output power estimation after partial generation shedding

Fig.11 shows the results of simulation, when some units in G1, G2, G3 are shed. The maximum error between the estimated values and the exact ones is less than 3 percent, and this result proves that the output power estimation method has a sufficient accuracy to be used with the stabilizing equipment.

Table 1 shows the results of simulation using the small power system. Number of units losing synchronism and that of units shed by stabilizing equipment are shown. For example, in case 1, if there is no stabilizing equipment, G1, G2 and G3 lose synchronism. On the other hand, if there is stabilizing equipment, all the generators are stabilized by shedding one unit out of four of G1 and four units out of six of G2. The following are shown from Table 1

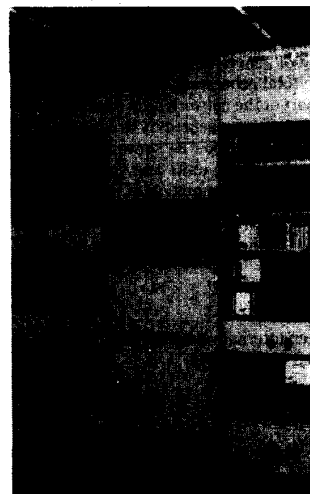


Fig.8 DEVELOPED STABILIZING EQUIPMENT

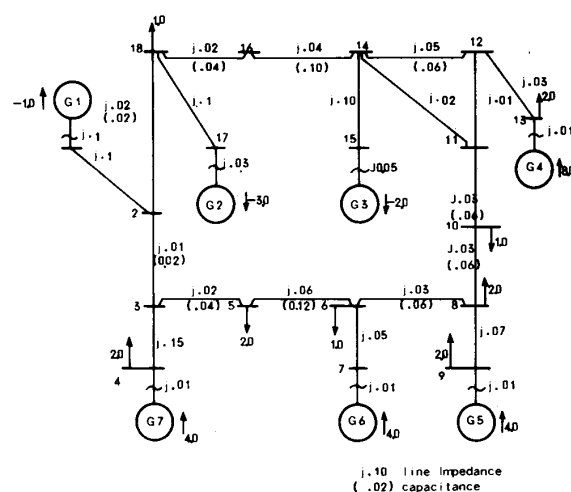


Fig.9 SMALL MODEL NETWORK

- (a) When the system is unstable, as in the case from 1 to 5, the stabilizing equipment takes precise actions to stabilize the system.
- (b) When the power system is stable as shown in case 6, the stabilizing equipment takes no control action. It is clear from these results that the proposed method has the capability to precisely predict system stability, and stabilize when the power system becomes unstable. The swing curves with and without the stabilizing equipment are shown in Fig.12.

(Example 2) Digital simulation for the TEPCO system.

The general configuration of TEPCO's power system is shown in Fig.1. The TEPCO system used in the following simulations consists of 73 generators, 380 nodes and 501 lines. There are 8 pumped-storage plants operating in pumping mode, and these plants consists of two to four units each. Three-phase faults which may occur at some place on the 500 KV transmission lines were investigated. Several hundred cases were studied.

As an example, the results of simulation for the fault location at seven points (F1 to F7) in Fig.1 are shown in Table 2. Table 2 compares the total capacity of generators which lose synchronism without the stabilizing equipment with that of generators to be shed by the stabilizing equipment.

Fig.13 shows the swing curves of the three pumped-storage plants (Y1, Y2 and Y3) located in the western area of the TEPCO system, and the reference generator (REFG) with and without the stabilizing equipment corresponding to case 4. In case 4, if there is no stabilizing equipment, all the pumped-storage plants except one plant located in the eastern area lose synchronism. On the other hand, if stabilizing equipment is used, the TEPCO system is stabilized by shedding a half unit of Y1 and Y2, and a quarter unit of Y3.

From these simulations, it was confirmed that the stabilizing equipment stabilizes TEPCO system by generator shedding of 10 to 30 percent of the total capacity of the generators which would lose synchronism if the stabilizing equipment was not used.

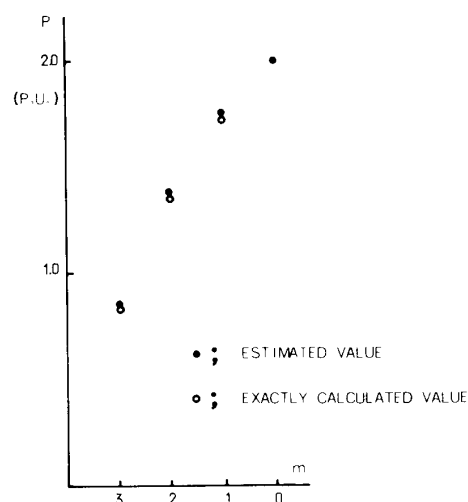


Fig.11 ACCURACY OF ESTIMATED POWER OUTPUT

Table 1. RESULT OF SIMULATION

Case NO.	Fault Location		Number of Units Lost or Shed					
			without stabilizing equipment			with stabilizing equipment		
	NF	NT	G1	G2	G3	G1	G2	G3
1	14	12	4	6	2	1	4	0
2	16	14	4	6	0	0	5	0
3	16	18	4	6	0	0	1	0
4	5	6	4	6	0	0	3	0
5	16	14	0	6	0	0	2	0
6	2	3	0	0	0	0	0	0

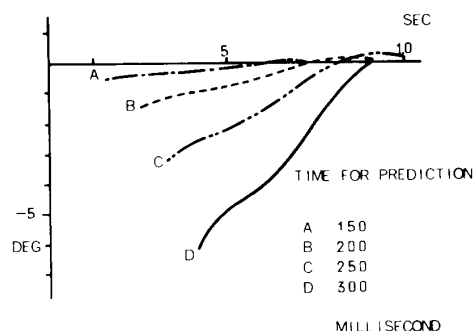


Fig.10 ACCURACY OF PREDICTED ANGLES

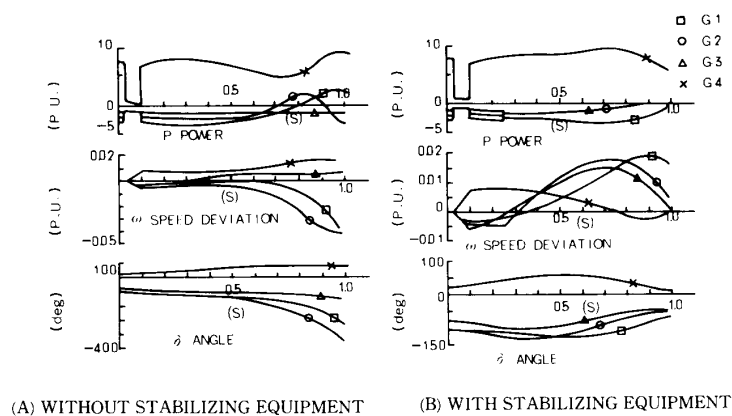
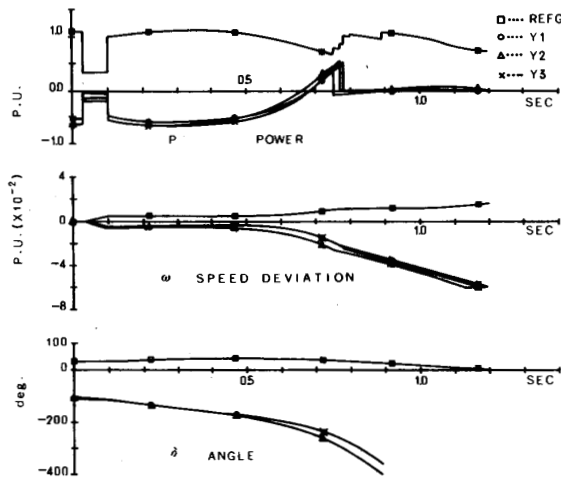
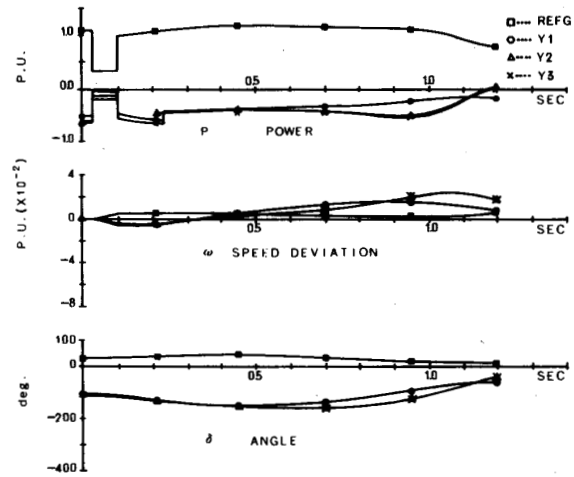


Fig.12 RESULT OF SIMULATION



(A) WITHOUT STABILIZING EQUIPMENT



(B) WITH STABILIZING EQUIPMENT

Fig.13 RESULT OF SIMULATION (TEPCO SYSTEM)

Table 2. RESULT OF SIMULATION (TEPCO SYSTEM)

Case NO.	Fault Location	Total output of generators	
		without stabilizing equipment	with stabilizing equipment
1	F 1	4190 (MW)	320 (MW)
2	F 2	4710	320
3	F 3	1830	640
4	F 4	4965	1210
5	F 5	0	0
6	F 6	1830	640
7	F 7	0	0

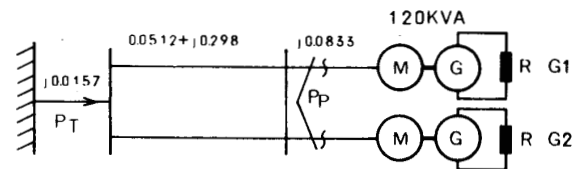


Fig.14 CONFIGURATION OF NETWORK IN ARTIFICIAL POWER SYSTEM

6. CONCLUSION

This paper presents a new method to prevent generators in pumped-storage plants from losing synchronism within one second after a disturbance occurrence. It also presents newly developed stabilizing equipment based on the proposed method. The following proposals are presented:

1. A method to estimate the angular velocity and phase angle of generators using the generator output power sampled at a rate of 600 times per second after disturbance occurrence.
2. A method to predict the swing of the generators during the next 200-300 milliseconds.
3. A method to predict the loss of synchronism by observing the relative phase angle difference between generators.
4. A method to estimate the minimum amount of pumped-storage capacity to be shed for stabilization.

As a result of installation of the newly developed stabilizing equipment, the amount of pumped-storage capacity to be shed for stabilization can be reduced to 10-30 percent of the total capacity of pumped storage plants which may be lost due to a major disturbance. This shows the equipment contributes to the enhancement of stability.

When every generator in the pumped-storage plants is in generating operation and some machine lose synchronism due to a major disturbance, the proposed method can be applied to minimize the loss of synchronism.

The developed stabilizing equipment was installed in the TEPCO power system in June, 1986.

5.2 Test by Artificial Power System

The stabilizing equipment was tested by experiment using an artificial power system to verify its effect. The configuration of model power system is shown in Fig.14. In Fig.14, two 120 KVA generator units operating in pumping mode are connected to the infinite bus through two circuits of 3.3 KV transmission lines. To simulate a pumped-storage unit, resistors are connected to the DC generator. The power consumed by the resistors is in proportion to the square of the induced voltage of the DC generator, and the induced voltage in turn is in proportion to rotor speed of the DC generator. Since the change of rotor speed during faults and a few hundred milliseconds after fault clearance is very small, the power consumed by the resistors is almost constant and it simulates the load of the pumped storage unit appropriately.

The generator constants at machine base are $H=8.3$, $X_d=X_q=1.42$, $X_d'=0.269$, $X_d''=X_q''=0.191$ and $T_{do}'=4.0$, respectively. System impedances at 100KVA base are also shown in Fig.14.

Fig.15 (A) shows the swing curves corresponding to the case without stabilizing equipment, two units lose synchronism on the first swing. On the other hand, Fig.15 (B) shows the swing curves corresponding to the case with stabilizing equipment. G2 is stabilized by shedding G1 at 0.2 second after fault occurrence.

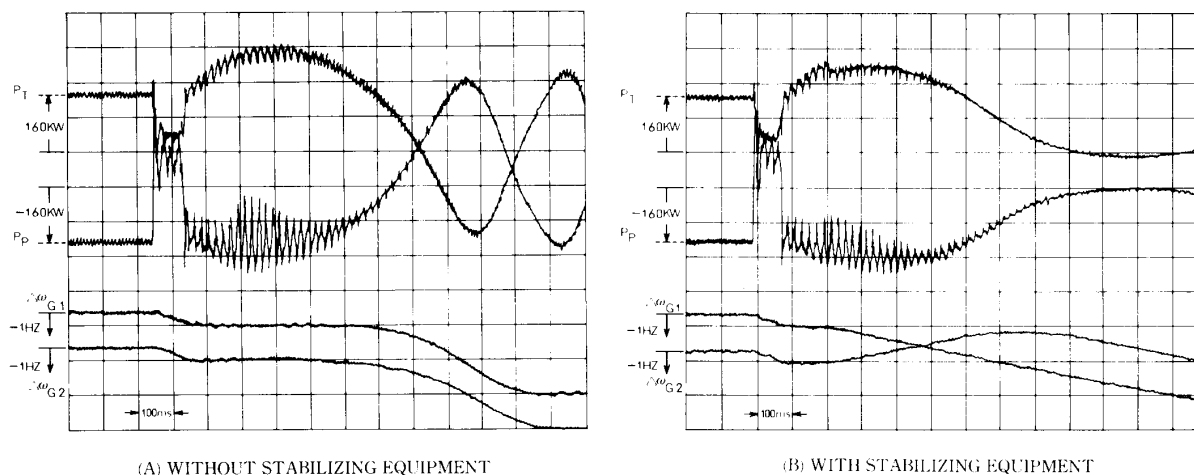


Fig.15 RESULTS OF TEST USING ARTIFICIAL POWER SYSTEM

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Discussion

John H. Doudna (Nebraska Public Power District, Columbus, NE): The authors have developed and implemented an interesting remedial action scheme which should improve system reliability and operating flexibility. Perhaps the most significant aspect of this remedial action scheme is that it is not just a theoretical abstract, but that it is a scheme which has actually been implemented. Once sufficient operating experience has been obtained, this discussor encourages the authors to produce a follow-up paper describing this operating experience.

The remedial action scheme presented by the authors was applied to pumped-storage plants to provide for unit shedding during pumping mode operation. This scheme controls the shedding of these units by comparing the phase angle, acceleration, and other attributes of the pumped-storage units to the same attributes of other generators which tend to be stable and operate as a coherent group. It is this discussor's observation that this scheme could be applied to any generating plant which is susceptible to instability and which can be compared against a group of stable units which operate as a coherent group. Could the authors offer their comments as to the applicability of their scheme to such situations, and do the authors have or foresee such an application on their system?

The scheme presented by the authors relies on independent processors located at each of their pumped-storage plants, and there is no communication between these processors. For a plant which is essentially isolated, such

a scheme seems appropriate. However, when several plants, such as Shintakase, Midono, and Azumi, are grouped together, rely on the same transmission lines and are collectively affected by the same system disturbance situations, it would appear necessary to have either a central processor to coordinate unit tripping among the several plants or some form of communication between the independent processors at each plant. Although each plant's independent processor could be set to coordinate with the processors at the other plants for ideal situations, what happens when the failure of a processor or other problems does not cause the required number of pumped-storage units at a particular plant to trip? How can the processor at one plant detect such occurrences at another plant? Could such situations lead to undertripping of units and result in system instability? Do the authors feel that some form of communication between the independent plant processors would enhance the scheme's ability to handle abnormal operating situations?

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M. Takahashi, K. Matsuzawa, M. Sato, K. Omata, R. Tsukui, T. Nakamura, and S. Mizuguchi: The authors wish to thank the discussor for his interest in this work and his comments on the paper.

As the discussor suggests, this scheme can be applied to any generating plant which may lose synchronism with the remaining generating plants which operate as a coherent group. Basically relative phase angle among

generators is used as the index which shows the degree of instability.

Some stabilizing equipments have been installed and some will be installed to TEPCO's power system, using an algorithm which is similar to that proposed in this paper.

Different phase angle criteria (threshold used in eq. (23)) were assigned to processors located at pumped-storage plants, respectively. The criteria were determined by off-line simulation studies in order to minimize the total capacity of units to be shed. Therefore, even if several plants, such as Shintakase, Midono, and Azumi, are grouped together and collectively affected by the same system disturbance situations, the total capacity of units to be shed becomes minimum.

Figure 3 in the paper shows the procedure to measure powers, estimate the angular velocity and phase angle, predict generator dynamics and loss of synchronism, decide the number of units to be shed, and trip the units. And each processor iterates the above procedure at intervals of Δt (20 ms) for

about 1 s after fault occurrence.

Therefore, even if the failure of a processor or other problem does not cause the required number of units at a particular plant to trip and, as a result, it induces the loss of synchronism of the other plants, the processor at each of the other plants will predict the loss of synchronism of the plant, decide the number of units to be tripped for stabilization, and trip them. By this mechanism, the extension of loss of synchronism to the remaining plants can be avoided.

We studied several ways to enhance the scheme's ability to handle abnormal operating situations. When the size of the stabilizing system including communication becomes larger, possibility of failure of the system increases, therefore the configuration of stabilizing equipment proposed in this paper was selected.

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